

What You Play, Where You Sit: Network Structure of Toxicity in Reddit Gaming Communities

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Abstract. Gaming communities are often compared by type, with genre and developer scale treated as risk factors that should inform moderation. We ask how far toxicity varies along those axes in nine Reddit gaming communities (550,641 posts and comments) spanning MMORPG and Soulslike genres, AAA/AA and indie developer scale, and a general-gaming baseline, scoring every record with two independent classifiers. Average toxicity does differ by community: Soulslike communities score above MMO communities under both classifiers, and the elevation is broad rather than category-specific, appearing across nearly every sub-category. The networks in which that toxicity sits, however, look much alike. Replies concord with the toxicity of the posts and comments they answer, more toxic threads are larger and more structurally viral, users mix assortatively on toxicity, and central users both produce and receive more of it, in all nine communities, under both classifiers, largely without reliable variation by genre, by developer scale, or against the baseline. Communities that differ in how much toxicity they carry are closely similar in how it is distributed across network position and mixing, which points moderation toward threads and positions rather than whole communities.

Keywords: online toxicity, social networks, assortative mixing, centrality, gaming communities

1 Introduction

Online gaming communities are large and inherently social spaces that many players treat as social lifelines [21]. Alongside advice, criticism, and norm negotiation, toxic behaviors remain persistent [22], harming well-being [8] and discouraging participation [15]; toxicity detection itself can also penalize minority speakers through biased annotation and classifiers [17].

A common working assumption is that toxicity is a property of community *type*. Players identify design features such as team interdependency, fairness, and communication affordances as triggers of toxic interaction [22], toxicity in stream chat differs by game genre [6], and visible in-community authority figures such as moderators shape which behaviors spread [18]. If community type is what matters, moderation should be tailored to it. This paper asks how far that assumption survives a multi-level test.

We analyze 550,641 posts and comments from nine Reddit gaming communities, scored with two independent toxicity detection models. Communities are selected along the two axes prior work argues should matter most: *genre* and *developer scale*. For genre, we contrast *MMOs* (massively multiplayer online role-playing games, or MMORPGs) with *Soulslikes*. MMOs are built around persistent shared worlds, whereas Soulslikes are difficult, combat-focused action games. For developer scale, we contrast *AAA/AA* titles with *indie* titles. AAA/AA titles come from large, well-resourced studios, whereas indie titles come from independent studios with smaller teams and budgets. We also include r/gaming as a general-gaming baseline. Together, these communities form a *sampling frame* accounting for between-community heterogeneity.

We examine toxicity at three levels and ask a question at each. **RQ1:** *How much do gaming communities differ in measured toxicity, and how is that difference distributed across categories of harmful language?* **RQ2:** *How is toxicity organized within discussion threads, in the concordance between adjacent comments and in the coupling between thread toxicity and cascade structure?* **RQ3:** *How does user toxicity relate to position in the reply network, through assortative mixing and through centrality-based exposure and production?* At each level we also test whether the pattern varies by genre, by developer scale, or from the r/gaming baseline; those tests are consolidated in Section 4.4.

2 Related Work

2.1 Community Type as a Proposed Determinant of Toxicity

Toxic comments are disrespectful, abusive, or unreasonable replies that tend to drive users out of a discussion [15]. Online play can support friendship and social capital [21], but analyses also reveal contested understandings of sexism in gaming discourse [12], and gaming cultures resist simple subcultural framing [14]. Two community-level attributes dominate comparative work. The first is genre: players attribute frustration and toxicity to design features that differ from game to game [22], and Twitch chats of shooter streams show a higher share of toxic messages than those of other genres [6]. In lab experiments, competition rather than violent content elevated short-term aggression [1]. This motivates our contrast between two genres that differ in how play is structured: MMORPGs, which emphasize persistent shared worlds and structured cooperation, and Soulslikes, which emphasize individual mastery and high difficulty. The second is developer scale: AAA/AA and indie titles differ in resources and in the distance between players and developers, and in-community authority shapes norms, with users

imitating moderators more than outsiders and responding to visible bans [18]; larger Twitch communities also show a higher share of toxic messages [6].

2.2 Toxicity in Interaction Networks

Prior work describes toxicity as spreading through a social structure [11], consistent with social contagion theory, which holds that behaviors and affective states spread through network ties [4]. Tree-based analyses of Reddit conversations find that a toxic comment raises the likelihood of subsequent toxic comments [20], aligning with work on how negative affect propagates through social systems [2]. Intercommunity conflict on Reddit is initiated by a small fraction of communities and marked by echo-chamber interaction [10], toxicity is homophilous in user networks [9], and a few highly central users can dominate discussion activity [5], with threads started by opinion leaders drawing more participants [16]. What this literature has not established is whether these regularities are stable across communities differing substantially in type [11].

3 Data and Methods

3.1 Communities and Corpus

We collected Reddit data using Arctic Shift [3], covering August 25 to October 15, 2025. Communities were selected on three criteria: the title maps unambiguously onto one genre and one developer scale; the subreddit is the primary community for that title; and it yields enough records in the window for thread- and user-level network estimation. This produced eight communities filling the four genre $\times$ scale combinations, plus r/gaming as a baseline (Table 1).

Table 1. The nine communities, with game title, genre, and developer scale as attributes; 550,641 records in total.

Subreddit	Game	Genre	Scale	Records
r/WoW	World of Warcraft	MMORPG	AAA/AA	99,947
r/ElderScrollsOnline	The Elder Scrolls Online	MMORPG	AAA/AA	55,214
r/Palia	Palia	MMORPG	Indie	80,136
r/AlbionOnline	Albion Online	MMORPG	Indie	25,263
r/DarkSouls	Dark Souls	Soulslike	AAA/AA	30,356
r/LiesofP	Lies of P	Soulslike	AAA/AA	50,994
r/HollowKnight	Hollow Knight	Soulslike	Indie	99,986
r/Ninesols	Nine Sols	Soulslike	Indie	8,777
r/gaming	—	General	Baseline	99,968

3.2 Preprocessing and Sampling

Text was normalized (Unicode NFKC, lowercased), with URLs, code spans, emphasis markup, and HTML entities removed and whitespace collapsed. We removed empty records, comments marked [deleted] or [removed], and records authored by [deleted] users, then harmonized post and comment schemas and enriched each record with parent and root context for thread reconstruction. A heuristic bot filter removed known bot and system accounts (e.g., AutoModerator), bot-flaired records, and records containing phrases such as “I am a bot.” Because three subreddits (r/WoW, r/HollowKnight, r/gaming) contained hundreds of thousands of records, we applied thread-preserving random sampling capped at approximately 100,000 records per subreddit: seed comments were sampled at random and retained with their full ancestry chain, so all sampled records remain embedded in complete, reconstructable threads.

3.3 Toxicity Measurement

Each record was annotated with two automated systems capturing different but complementary forms of harmful language; when body text was unavailable, scoring was applied to the post title (1.18% of records). The OpenAI **omni-moderation-latest** model [13] evaluates text across categories including harassment, hate, self-harm, sexual content, violence, and illicit activity, with severity-specific subcategories and continuous per-category confidence scores. Detoxify **original-small** [7], trained on Jigsaw benchmark data, outputs an overall toxicity probability plus severe toxicity, obscenity, threat, insult, and identity attack.

Because the OpenAI API provides no single continuous measure of overall harmfulness, we constructed a scalar proxy by taking the maximum confidence score across all moderation categories. These scores are model confidence rather than a calibrated severity scale, so the aggregation is a pragmatic summary. This score and Detoxify’s toxicity probability are the primary measures throughout, and we use *toxicity* as shorthand for harmful language as operationalized by these models. The two show moderate alignment on this corpus (Spearman’s $\rho = .476$, Pearson’s $r = .455$, both $p < .001$).

We report continuous scores throughout. As a robustness check we replicated every analysis with binary indicators, using OpenAI’s flagged indicator and Detoxify thresholded at the quantile matching OpenAI’s flag prevalence (approximately 6.36%), with sensitivity analyses at $\pm 5\%$; these reproduce the direction of every network-level finding below and the broad pattern of Soulslike communities scoring higher across most subcategories.

3.4 Thread Trees and User-Reply Networks

We constructed reply trees at the post-comment level, in which nodes are posts or comments and directed edges connect a parent to each direct reply, using Reddit’s comment-parent identifiers. Only replies whose parent was present in the dataset were retained, so each connected component corresponds to an observed

thread, rooted at the original post when available. From the same structures, we built one directed user-reply network per subreddit, in which nodes are unique accounts and an edge runs from a commenting user to the author of the post or comment they replied to; self-replies were excluded. Bots and [deleted] authors were excluded from the networks.

4 Results

4.1 Community-Level Differences (RQ1)

Figure 1 reports mean toxicity per community. The models differ in absolute scale, with Detoxify assigning higher means throughout, but agree at the extremes: r/LiesofP and r/DarkSouls rank highest under both measures, r/Palia and r/ElderScrollsOnline lowest. Agreement is weaker in the middle, where r/gaming and r/WoW rank higher under Detoxify (.097, .085) than under OpenAI (.050, .048). Under both classifiers, all four Soulslike communities have higher mean toxicity than all four MMO communities, whereas developer scale shows no such separation. The differences are modest: genre accounts for at most 2.2% of record-level variance and developer scale for under 0.5%.

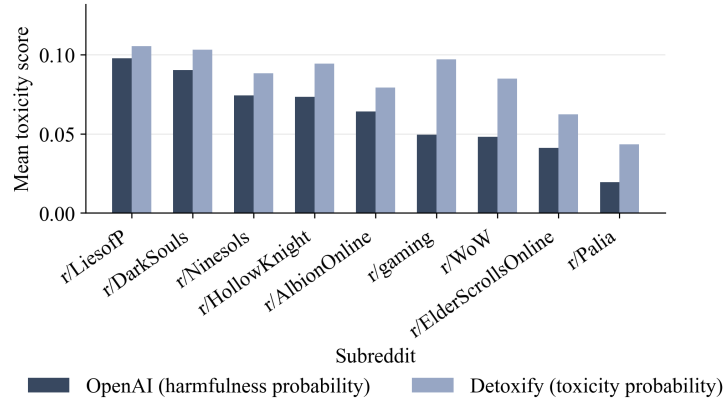


Fig. 1. Mean toxicity by community under OpenAI and Detoxify.

How the difference is distributed across categories. Table 2 decomposes mean scores by subcategory. Soulslike communities score higher than MMO communities on ten of the eleven subcategories, in both the AAA/AA and the indie comparison. Violence moves most (3.5× in the AAA/AA cell, 2.7× in indie) and the other OpenAI content-topic categories move with it (except illicit activity in the indie comparison); Detoxify shows the same broad pattern through

its categories, with insult $1.7\text{--}1.9\times$ and obscenity $1.5\text{--}2.1\times$ higher in Soulslike communities. The elevation we observe is therefore general rather than category-specific: nearly everything the classifiers score is somewhat more present in Soulslike communities, and no category separates the genres by enough to indicate a distinct *kind* of harm. Hate is the single exception, running slightly lower in Soulslike communities in both comparisons.

Table 2. Mean subcategory scores by genre $\times$ developer scale, rounded to four decimals. OpenAI values are means over severity subcategories; Detoxify categories are reported directly. Ratios given in the text are computed before rounding.

Category	AAA/AA		Indie	
	MMO	Soulslike	MMO	Soulslike
<i>OpenAI</i>				
Harassment	.0123	.0139	.0104	.0124
Hate	.0008	.0006	.0008	.0007
Illicit activity	.0028	.0087	.0047	.0048
Self-harm	.0007	.0017	.0008	.0017
Sexual content	.0016	.0034	.0017	.0036
Violence	.0104	.0360	.0097	.0261
<i>Detoxify</i>				
Severe toxicity	.0009	.0021	.0007	.0019
Obscene	.0310	.0455	.0197	.0418
Threat	.0004	.0010	.0006	.0007
Insult	.0113	.0192	.0089	.0166
Identity attack	.0005	.0007	.0005	.0006

4.2 Thread-Level Concordance and Cascade Coupling (RQ2)

Turning to thread structure, we ask whether toxicity in replies covaries with the toxicity of the comments they respond to, estimating for each subreddit a mixed-effects model at the reply-edge level, $\text{child_tox}_e = \beta_0 + \beta_1 \text{parent_tox}_e + u_{\text{thread}} + \varepsilon_e$, where each observation is a parent-reply pair and u_{thread} is a random intercept capturing within-thread dependence. We call β_1 local *concordance* rather than contagion: the design is cross-sectional, and influence and homophily are generically confounded in observational network data [19].

Across all nine subreddits, β_1 was positive for both measures (Figure 2, top). This was robust to inference method: both model-based p -values and within-thread permutation tests, which shuffle toxicity among nodes within each thread while preserving reply structure, indicated that the observed slopes were unlikely under random assignment.

To assess whether more toxic discussions are deeper or larger, we computed per-thread cascade metrics: thread size, maximum reply depth, and structural

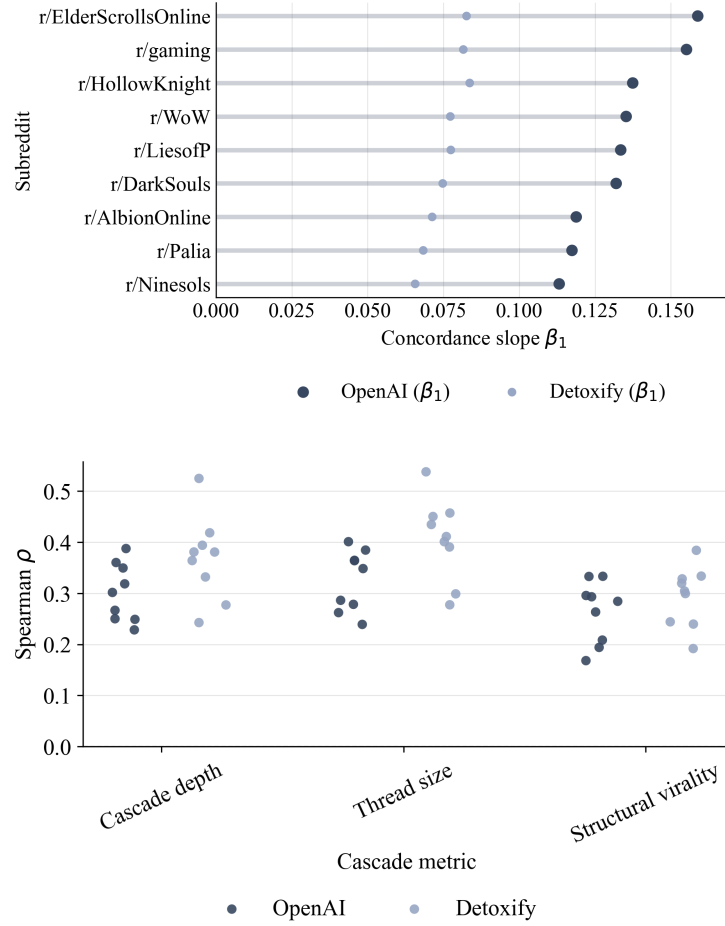


Fig. 2. Thread-level structure, one point per community, under both classifiers. Top: parent-to-reply concordance β_1 . Bottom: Spearman correlations between mean thread toxicity and cascade depth, size, and structural virality.

virality (average shortest-path distance between node pairs), the last estimated via pair-sampling on the full thread subgraph to avoid the downward bias of induced-subgraph node sampling. Across subreddits and both measures, mean thread toxicity was positively rank-associated with all three, and this held under permutation testing (Figure 2, bottom): more toxic threads tend, on average, to be larger, deeper, and more structurally viral than less toxic threads. These associations were consistent in direction across communities and cascade metrics, though their magnitudes varied by metric and classifier.

4.3 User-Level Position and Mixing (RQ3)

At the user level we aggregated toxicity as each user’s mean score across their posts and comments, then quantified assortativity as the Pearson correlation between the toxicity of source and target across reply ties, $r = \text{corr}(\text{tox}(u), \text{tox}(v))$ for edges $u \rightarrow v$. Significance was assessed with permutation tests preserving reply structure while shuffling toxicity across users, with an analytic Fisher- z approximation as a backup when the permutation null collapsed.

All nine subreddits exhibited positive assortativity under both models, ranging from $r = .0566$ (r/Palia) to $r = .1182$ (r/Ninesols) under OpenAI and from $r = .0369$ (r/ElderScrollsOnline) to $r = .0654$ (r/HollowKnight) under Detoxify (Figure 3). While OpenAI consistently produced larger r than Detoxify, the direction of effects was uniform. Permutation tests indicated these values were unlikely to arise from random mixing, supporting the conclusion that users with similar toxicity levels tend to reply to one another more than expected by chance. In the three communities where the null was undefined (r/ElderScrollsOnline, r/Palia, r/DarkSouls), Fisher- z tests still gave evidence of non-zero correlations ($p < .001$).

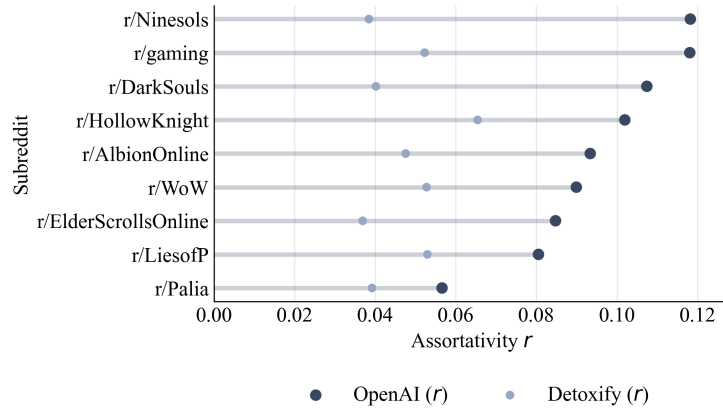


Fig. 3. User-level toxicity assortativity (r) by community under OpenAI and Detoxify.

Next, we investigated whether where a user sits in the reply network relates to the level of toxicity they produce and receive. We computed four node-level centrality measures on each community’s directed user-reply graph, in-degree, out-degree, betweenness, and eigenvector centrality, and two user-level toxicity summaries per model: produced toxicity (mean toxicity of a user’s authored content) and received toxicity (mean toxicity of replies received). Associations were measured with Spearman’s ρ and assessed with node-label permutation tests.

Across all nine subreddits, all four centrality measures, both toxicity directions, and both models, the correlations were positive, ranging from $\rho = .133$ to $.459$, with all two-sided permutation p -values below $.05$ (Figure 4). Strength depended on centrality type and on toxicity direction, and the dependence is the same under both classifiers: in-degree, betweenness, and eigenvector centrality correlate more strongly with received toxicity, while out-degree shows the reverse, and the gap is widest for in-degree. This asymmetric pattern suggests a distinction between network positions associated with exposure to toxicity and those associated with its expression: users who receive many replies are disproportionately exposed to toxic ones, and users who reply to many others are more likely to produce them.

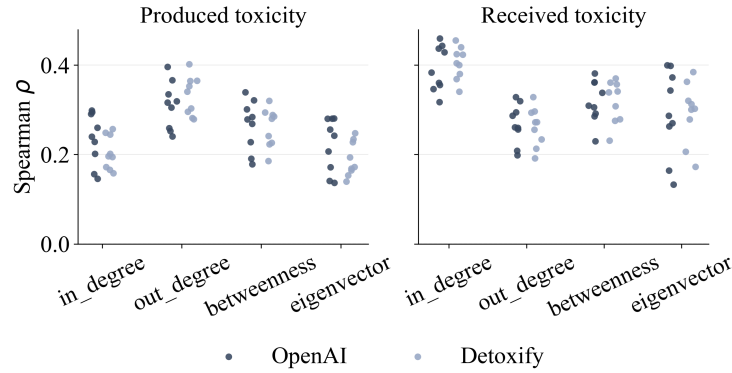


Fig. 4. Distribution of centrality–toxicity correlations across communities.

4.4 Variation by Community Type

The regularities above were estimated separately in nine communities differing in genre and developer scale. At every level we tested whether their magnitude varied with community type, using label-permutation tests on subreddit assignments and Holm-corrected comparisons against the r/gaming baseline. For concordance slopes, neither the four-cell comparison nor the genre $\times$ scale interaction was significant for either classifier (all $p > .05$). Cascade coupling was indistinguishable from chance across the four cells for most metrics, the exception being Detoxify’s toxicity–virality association (four-cell $p = .018$) alongside a clearly non-significant interaction ($p = .94$). For assortativity no group contrast was significant and no group differed from r/gaming under either model (all $p \geq .34$; Holm-corrected $p = 1.0$). For centrality, one isolated contrast reached $p = .03$ (Detoxify, produced toxicity, out-degree), and no group differed from r/gaming after Holm correction.

Differences from the r/gaming baseline are confined to a few group–measure pairs. Under Detoxify, Soulslike indie communities showed weaker toxicity-cascade coupling than r/gaming for depth ($\rho = .313$ vs. $.431$) and virality ($.257$ vs. $.371$), with virality also weaker under OpenAI ($.205$ vs. $.387$; all $p = .029$). MMO indie communities showed stronger centrality-received-toxicity associations at uncorrected thresholds, none surviving Holm correction.

5 Discussion

Average toxicity differs across community types, modestly and broadly: Soulslike communities score above MMO communities under both classifiers, with the elevation running across nearly every subcategory rather than concentrating in one form of harm. Where the communities do not differ is in position and mixing. Every network-level regularity we measure holds in all nine communities, in the same direction, under both classifiers, largely without reliable variation by genre, developer scale, or baseline. The axes along which comparative work expects toxicity to vary track differences in how much toxicity a community carries, but not in how it is distributed across positions and ties.

For moderation practice, this shifts the target. Sorting communities by genre, by studio scale, or by aggregate toxicity score has little to work with here: the relevant variance is not between communities but within them. Interventions aimed at reply-level escalation, at threads whose cascade structure is coupled to their toxicity, and at positions that concentrate exposure are supported by patterns holding in every community we examined. The centrality asymmetry is the most directly actionable: high in-degree users are candidates for protection and high out-degree users for monitoring, and these are different populations needing different responses. The results also caution against treating any single community or genre as inherently toxic; genre might be associated with the frequency of common hostile language but not with its distribution across the network.

We expect the direction of the network regularities, more than their magnitudes, to be more likely to hold in other threaded Reddit communities, since they recur across the varied communities here and parallel patterns reported outside gaming [9,20]. The between-community differences, resting on two titles per cell in one seven-week window, should not be generalized to genres or developer scales.

5.1 Limitations

The design is cross-sectional: concordance between adjacent comments is consistent with influence, selection, or shared context, and observational network data cannot separate them without strong assumptions [19], so we make no causal claim. The sampling frame has two titles per genre $\times$ scale cell, confounding those attributes with era, community age, size, and moderation regime. The genres may also overlap: Dark Souls includes online co-op and invasions, and

may share some audience with *The Elder Scrolls Online*. We chose titles to fill the cells under the criteria in Section 3.1, but other qualifying titles could have been chosen instead and might yield different results. We rely on automated classifiers that are neither tailored to nor manually validated on gaming discourse: they may misread platform-specific slang, emotes, and sarcasm [8], exhibit racial biases where profanity or reclaimed slurs are present [17], and score records outside conversational context. Because the OpenAI summary score is a maximum that includes violence and other content categories, some flagged language may describe game content (combat, death, bosses) rather than hostile interaction; this bears most on the genre contrast, while the network findings also hold under Detoxify, whose categories target interpersonal abuse. Future work should qualitatively and thoroughly inspect how different types of posts are scored. Finally, betweenness and eigenvector centrality may be unstable under the edge truncation; deleted content and moderation actions are unobserved.

Ethical considerations. This study analyzes publicly available Reddit posts and comments; we did not interact with users or intervene in community activity. To limit re-identification risks we report only aggregate results, reproduce no usernames, and quote no individual comments, and we treat model outputs as proxies for harmful language rather than ground truth.

AI tools disclosure. The authors used ChatGPT and Claude for text refinement and code review, reviewed and edited the content, and take full responsibility for it.

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